# Configuring STM32WB for Bluetooth LE without an LSE crystal

**AME MCU Support Center JT**
Associate III

on 2021-11-01 12:05 PM
- edited on 2025-08-01 5:08 AM by ADMIN **Laurids_PETERSEN**

## Version history

**Last update:**
2025-08-01 5:08 AM

**Updated by:**
ADMIN **Laurids_PETERSEN**

Feedback

## Summary

This article summarizes a method to remove the external 32.768 LSE crystal and replace its functionality with a clock reference sourcing from the 32MHz HSE.  It also describes the

## Contributors

necessary configuration modifications to implement a Bluetooth LE firmware project to run without the external LSE crystal and run from the HSE clock.

Laurids_PETERSEN

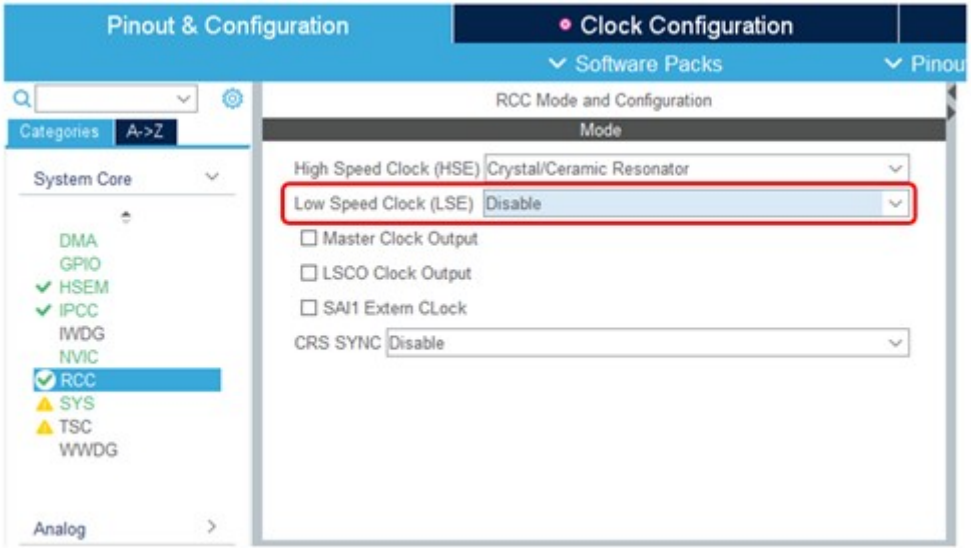
## Overview

The Bluetooth Low Energy specification requires that a low frequency crystal, not exceed a frequency deviation of 500 ppm. On the STM32WB, this low frequency functionality is provided by the external LSE crystal (typically). The LSE serves as the clock reference for the RF wake up subsystem of the BLE radio. When the main system clocks are shutoff during low power mode, it will remain running during this event if low power modes, such as STOP/STOP2, have been enabled. This is unlike the operation of the rest of the system clocks generated by the 32MHz system clock. As the radio is required to wake up precisely on time to service the connection intervals, thus it always requires an active clock source.  This functionality is typically provided by the LSE clock. Furthermore, the LSE is also used to clock the RTC for application timing, as it is an accurate and stable clock source compared to the internal LSI oscillators.
It is possible to remove the external LSE crystal and use the 32MHz HSE in its place, which is already used as the main system clock source. The tradeoff with this configuration is that of no longer being able to go into low power modes lower than SLEEP mode. For example, STOP/STOP2/STANDBY modes would no longer be available to the user, since the HSE is inevitably shutoff when the system enters these deeper low power modes.  In a system implementation where there is no low power requirement, such as devices that are mains powered (not battery powered), this may be a viable alternative. If your application is ok with not supporting low power modes lower than SLEEP mode, then the following steps can be followed to configure the STM32WB program to run only from the HSE only and not use the LSE.
Note that all examples in the STM32CubeWB firmware package are preconfigured to use the

Feedback

LSE for the RF wake up and RTC, so any can be used to demonstrate the changes that need to be made for this. In this case, the “BLE_p2pServer” example, found under “STM32Cube_FW_WB_V1.12.0\Projects\P-NUCLEO-WB55.Nucleo\Applications\BLE\BLE_p2pServer”, will be used to demonstrate how to do this. The changes will be applied from its respective CubeMX project.

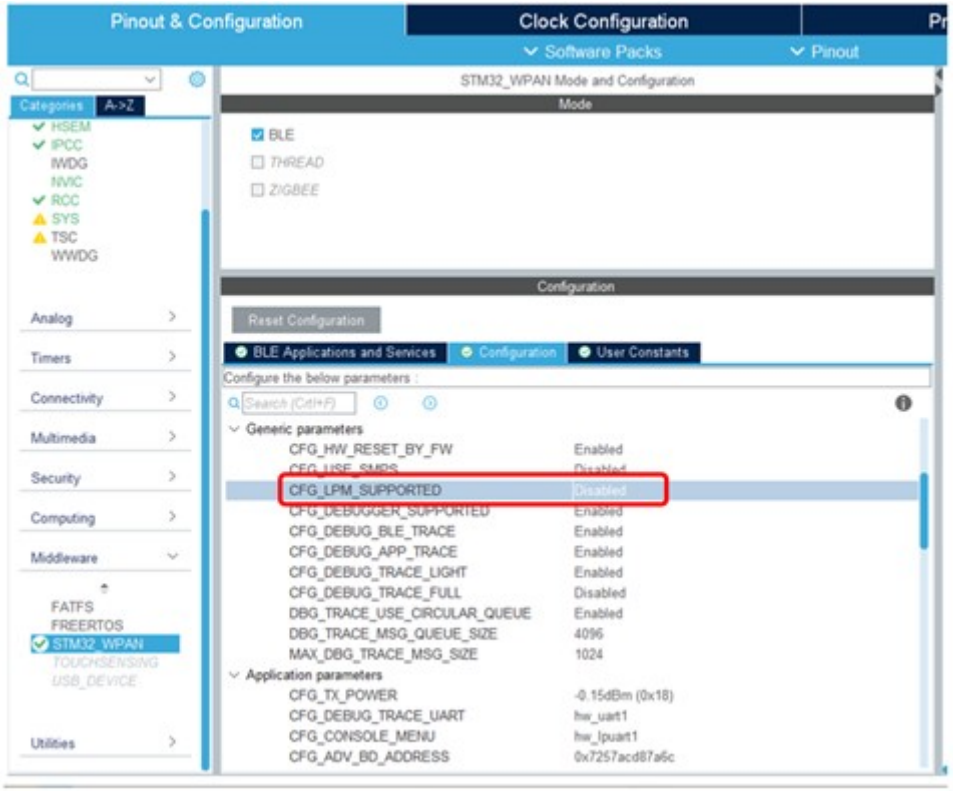
1. **Open its BLE_p2pServer.ioc with CubeMX v6.3.0+.**
2. **From the “RCC” configuration window, change the LSE mode to “Disable”**




3. **From the “STM32_WPAN” middleware configuration window, go to the “Configuration” tab**

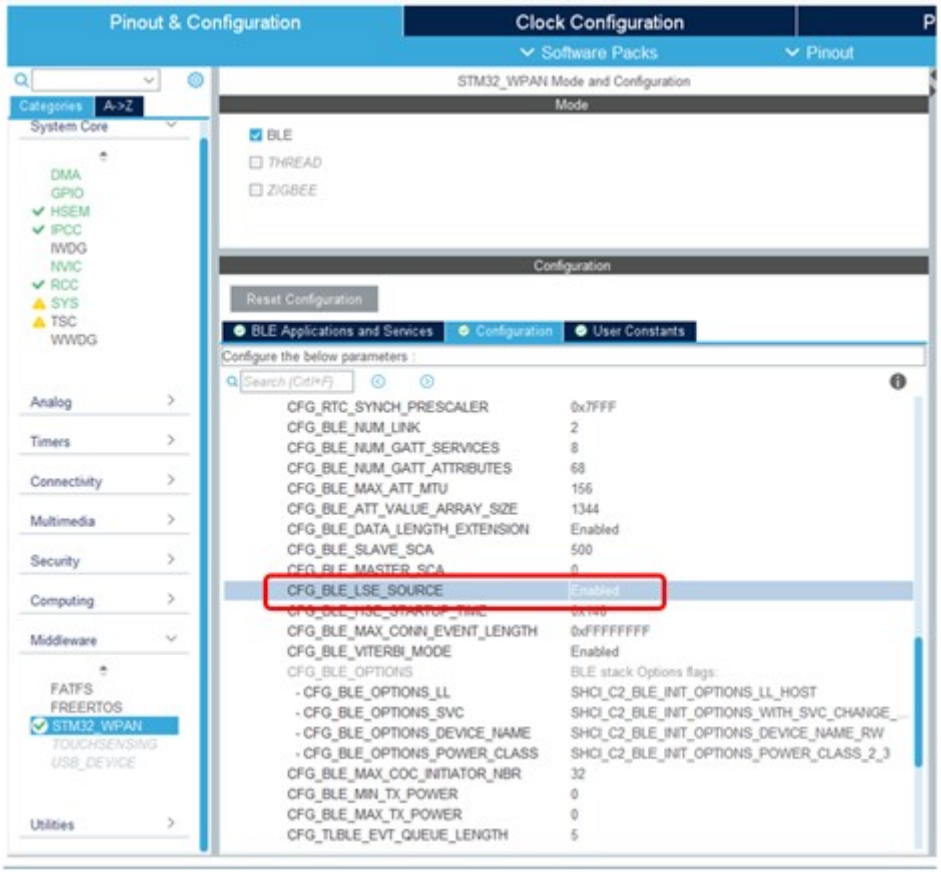
Feedback

a. Change the CFG_LPM_SUPPORTED parameter to “Disable”. This disables the low power manager support to keep the system always in Run mode.



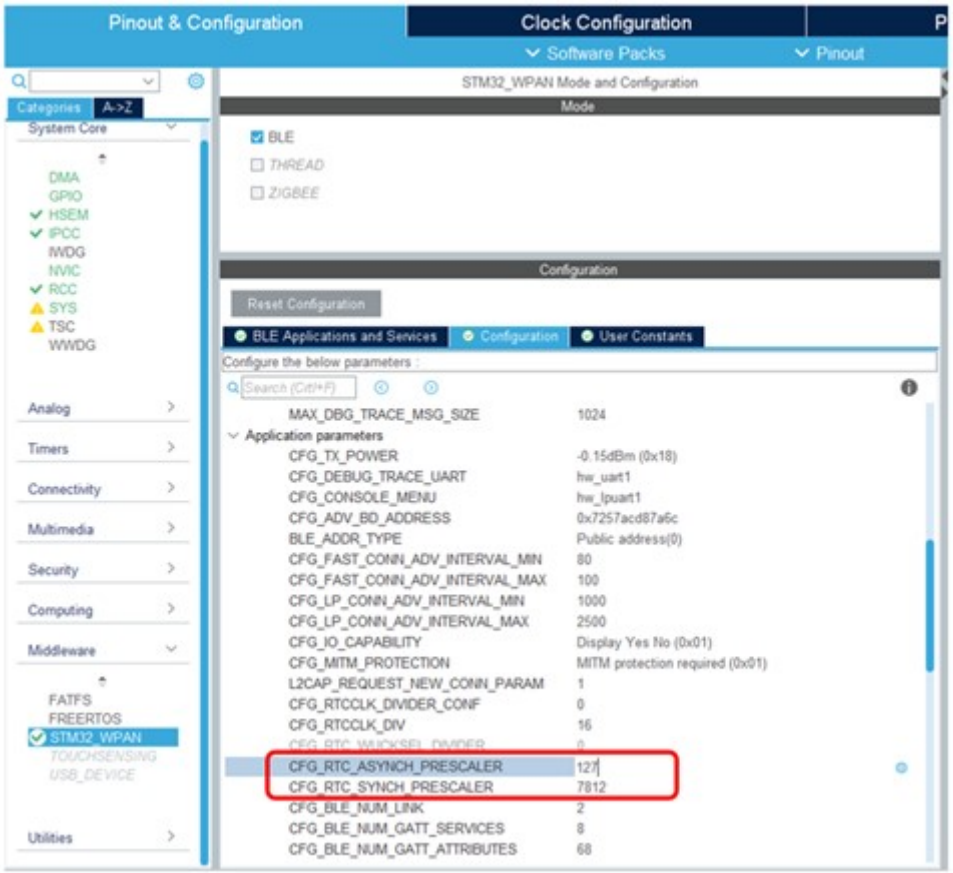

b. Change the CFG_BLE_LSE_SOURCE parameter to “Enable” to select the extern HSE as the source of the RF wake-up.

Feedback




c. Change the CFG_RTC_ASYNCH_PRESCALER parameter to 127 and CFG_RTC_SYNCH_PRESCALER to 7812 to obtain an internal clock frequency of 1 Hz (ck_spre). This will also keep the same real time base for the Virtual Timer utility (from hw_timerserver.c).

Feedback

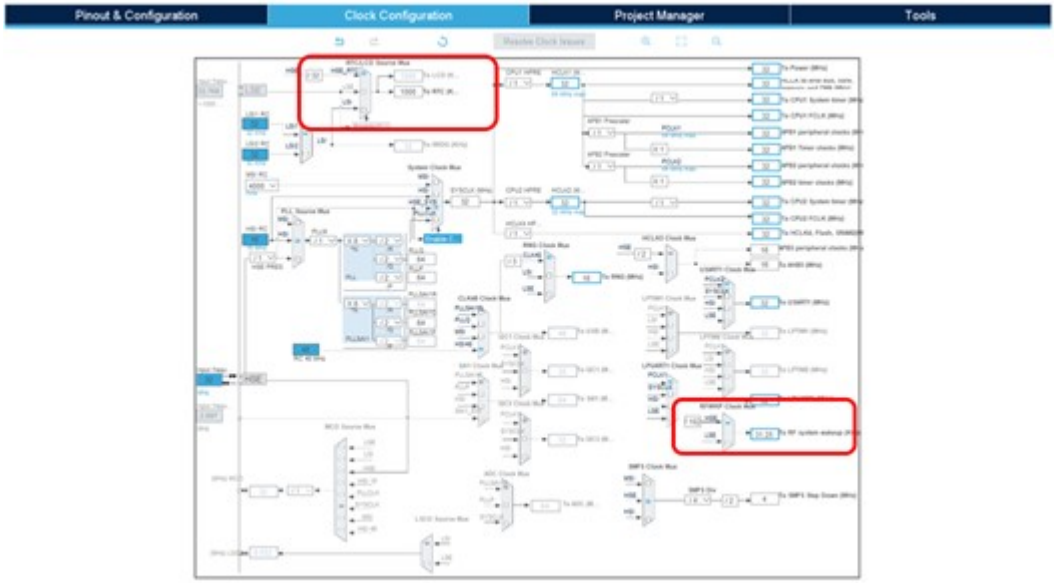


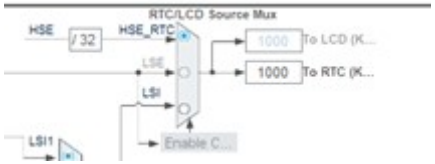

4. Change the clock tree configuration

    a. Select “HSE_RTC” for the RTC Source Mux

    b. Select “HSE” for the RF system wakeup

Feedback



5. Regenerate the code project for your favorite IDE.

6. Once code project is opened in the IDE, find app_conf.h to make the following manual edit to the #define CFG_TS_TICK_VAL. Note that this parameter is not available within CubeMX as it is considered part of the example's application user code.

Feedback

```
/** tick timer value in us */
//#define CFG_TS_TICK_VAL          DIVR( (CFG_RTCCLK_DIV * 1000000), LSE_VALUE )
#define CFG_TS_TICK_VAL          DIVR( (CFG_RTCCLK_DIV * 1000000), HSE_VALUE/32 )
```

7. Find stm32wbxx_hal_msp.c and add the following lines of code inside HAL_RTC_MspInit().

```
/**
 * @brief RTC MSP Initialization
 * This function configures the hardware resources used in this example
 * @param hrtc: RTC handle pointer
 * @retval None
 */
void HAL_RTC_MspInit(RTC_HandleTypeDef* hrtc)
{
  if(hrtc->Instance==RTC)
  {
  /* USER CODE BEGIN RTC_MspInit 0 */

    HAL_PWR_EnableBkUpAccess(); /**< Enable access to the RTC registers */

    /**
    * Write twice the value to flush the APB-AHB bridge
    * This bit shall be written in the register before writing the next one
    */
    HAL_PWR_EnableBkUpAccess();

    __HAL_RCC_RTC_CONFIG(RCC_RTCCLKSOURCE_HSE_DIV32); /**< Select HSE/32 as RTC Input

  /* USER CODE END RTC_MspInit 0 */
    /* Peripheral clock enable */
    __HAL_RCC_RTC_ENABLE();
    __HAL_RCC_RTCAPB_CLK_ENABLE();
    /* RTC interrupt Init */
    HAL_NVIC_SetPriority(RTC_WKUP_IRQn, 0, 0);
    HAL_NVIC_EnableIRQ(RTC_WKUP_IRQn);
  /* USER CODE BEGIN RTC_MspInit 1 */
  HAL_RTCEx_EnableBypassShadow(hrtc);
  /* USER CODE END RTC_MspInit 1 */
  }

}
```

8. Build project and run on target (NUCLEO-WB55RG). At this point, the STM32WB is no longer utilizing the LSE.

Feedback

BLE | STM32 MCU Products | STM32CubeMX | STM32WB series

2 Kudos

# COMMENTS

**gongw**
Associate II

2022-01-20 1:43 AM

Hi JT,

For CubeMX6.4.0 and CubeWBv1.13.1, there are few differences:

step 6:

/** tick timer values */

//#define CFG_TS_TICK_VAL DIVR( (CFG_RTCCLK_DIV * 1000000), LSE_VALUE )

//#define CFG_TS_TICK_VAL_PS DIVR( ((uint64_t)CFG_RTCCLK_DIV * 1e12), (uint64_t)LSE_VALUE )

#define CFG_TS_TICK_VAL DIVR( (CFG_RTCCLK_DIV * 1000000), HSE_VALUE/32 )

#define CFG_TS_TICK_VAL_PS DIVR( ((uint64_t)CFG_RTCCLK_DIV * 1e12), (uint64_t)(HSE_VALUE/32) )

Feedback

step 7: CubeWBv1.13.1 has already configure the RTCClockSelection as RCC_RTCCLKSOURCE_LSE

the regenerated coded by CubeMXv6.4.0 has already configure the RTCClockSelection as RCC_RTCCLKSOURCE_HSE_DIV32

```
 /** Initializes the peripherals clock
 */
  PeriphClkInitStruct.PeriphClockSelection = RCC_PERIPHCLK_RTC;
  PeriphClkInitStruct.RTCClockSelection = RCC_RTCCLKSOURCE_HSE_DIV32;
  if (HAL_RCCEx_PeriphCLKConfig(&PeriphClkInitStruct) != HAL_OK)
  {
   Error_Handler();
  }
```

Br,

gongw

0 Kudos

Feedback

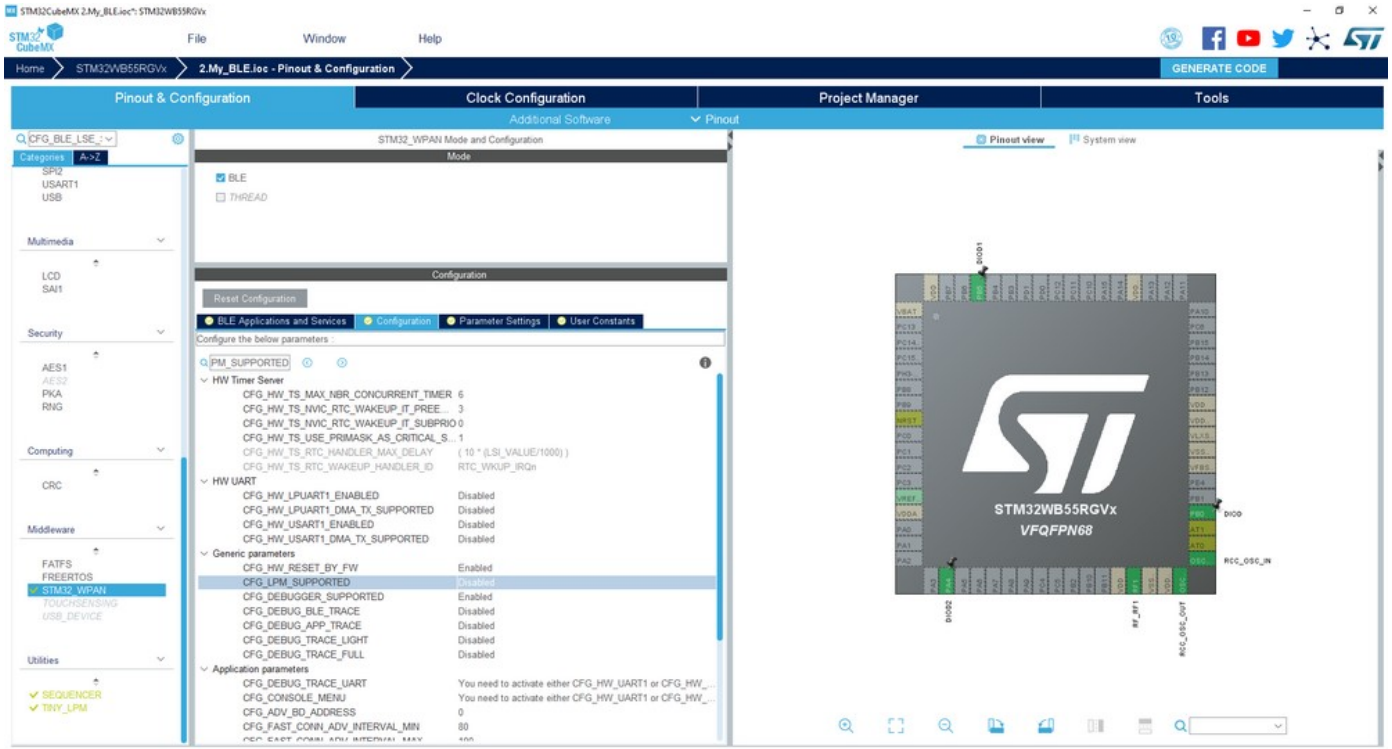
**lRakh.1**
Associate III

2022-10-11 4:05 PM

Hi, thank you so much for your answer.

Can you a little clarity about - ***CFG_BLE_LSE_SOURCE and CFG_RTC_ASYNCH_PRESCALER***

I don't have this setting in my CubeMX, please see the attached image. Do you know how I can fix it




?

Feedback

0 Kudos

**gongw**
Associate II

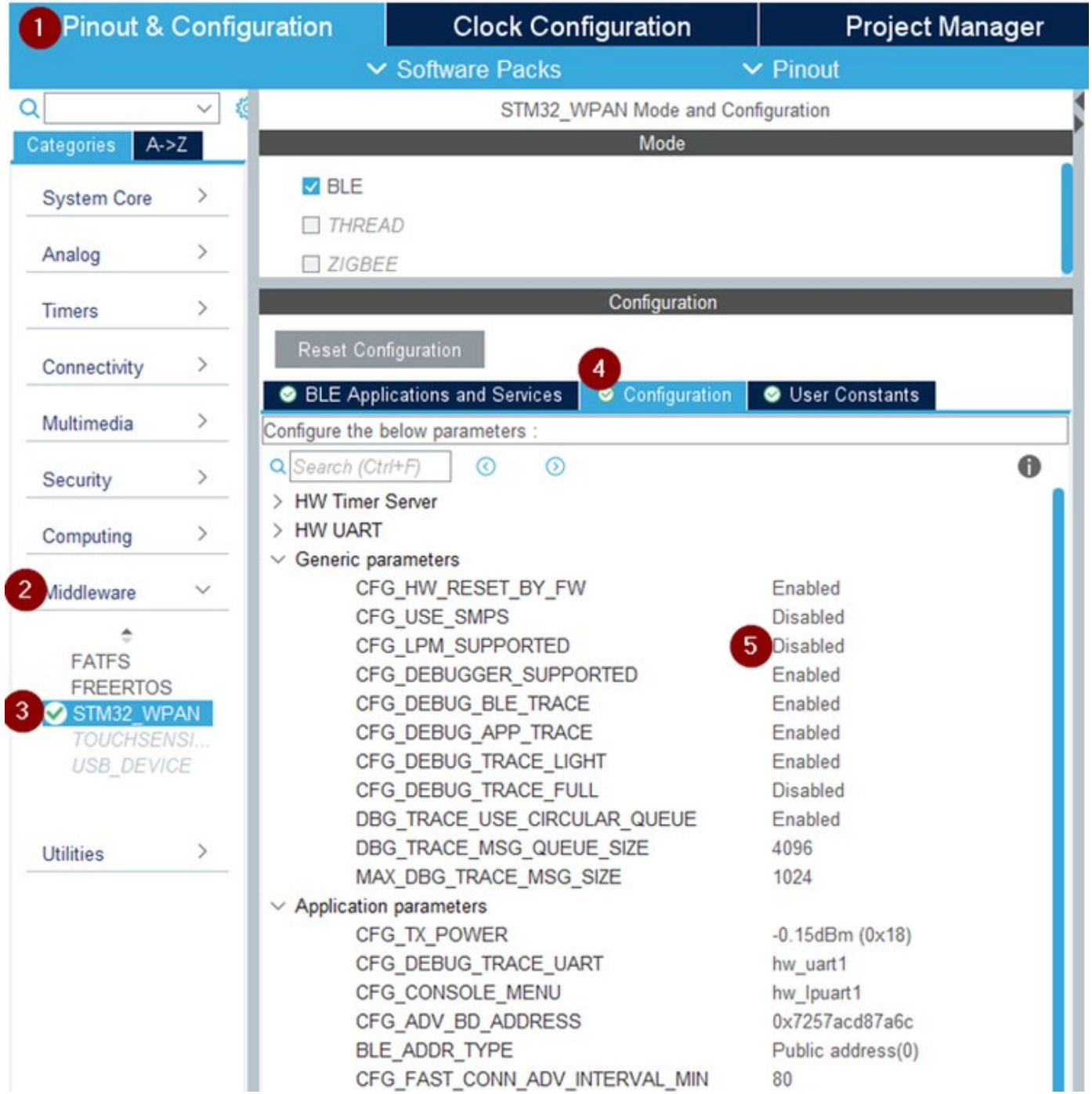
2022-10-11 10:09 PM

Hi,

First please use the latest CubeMX v6.6.1 and CubeWBv1.14.1.

Please go down to the middle position of Application parameters by scrolling the lightblue scrollbar

Feedback


1 Pinout & Configuration
Clock Configuration
Project Manager
Software Packs
Pinout
Categories
A->Z
System Core
Analog
Timers
Connectivity
Multimedia
Security
Computing
2 Middleware
FATFS
FREERTOS
3 STM32_WPAN
TOUCHSENSI...
USB_DEVICE
Utilities
STM32_WPAN Mode and Configuration
Mode
BLE
THREAD
ZIGBEE
Configuration
Reset Configuration
4
BLE Applications and Services
Configuration
User Constants
Configure the below parameters :
Search (Ctrl+F)
HW Timer Server
HW UART
Generic parameters
CFG_HW_RESET_BY_FW Enabled
CFG_USE_SMPS Disabled
CFG_LPM_SUPPORTED 5 Disabled
CFG_DEBUGGER_SUPPORTED Enabled
CFG_DEBUG_BLE_TRACE Enabled
CFG_DEBUG_APP_TRACE Enabled
CFG_DEBUG_TRACE_LIGHT Enabled
CFG_DEBUG_TRACE_FULL Disabled
DBG_TRACE_USE_CIRCULAR_QUEUE Enabled
DBG_TRACE_MSG_QUEUE_SIZE 4096
MAX_DBG_TRACE_MSG_SIZE 1024
Application parameters
CFG_TX_POWER -0.15dBm (0x18)
CFG_DEBUG_TRACE_UART hw_uart1
CFG_CONSOLE_MENU hw_lpuart1
CFG_ADV_BD_ADDRESS 0x7257acd87a6c
BLE_ADDR_TYPE Public address(0)
CFG_FAST_CONN_ADV_INTERVAL_MIN 80

Feedback

Br,

willee

0 Kudos

**lRakh.1**
Associate III

2022-10-15 1:38 PM

Hi Laura, thank you for your response, I did your recommendation, but still in my CubeMX don't have settings for CFG_BLE_LSE_SOURCE

if I correctly understand I can use the command in the code to fix it, how do you think?

#define CFG_BLE_LSE_SOURCE 1

and CubeWBv1.14.1 is it only examples of code, not more, yes?

0 Kudos

**gongw**
Associate II

2022-10-16 5:45 AM

Feedback

Hi,

For settings for CFG_BLE_LSE_SOURCE, you need to modify CFG_BLE_LSE_SOURCE as below after generated code.

```
/**
 * LsSource
 * Some information for Low speed clock mapped in bits field
 * - bit 0:   1: Calibration for the RF system wakeup clock source   0: No calibration for the RF system wakeup clock source
 * - bit 1:   1: STM32W5M Module device                              0: Other devices as STM32WBxx SOC, STM32WB1M module
 */
#if defined(STM32WB5Mxx)
  #define CFG_BLE_LSE_SOURCE  (SHCI_C2_BLE_INIT_CFG_BLE_LSE_NOCALIB | SHCI_C2_BLE_INIT_CFG_BLE_LSE_MOD5MM_DEV)
#else
  #define CFG_BLE_LSE_SOURCE  (SHCI_C2_BLE_INIT_CFG_BLE_LSE_CALIB | SHCI_C2_BLE_INIT_CFG_BLE_LSE_OTHER_DEV)
#endif
```

Br,

willee

0 Kudos

Feedback

**Mikisoft**
Associate

2025-01-08 2:27 PM

Thank you! This is exactly what I needed.

0 Kudos

**Adrian Adamson**
Senior

2025-10-03 8:22 AM

I'm trying this now with STM32CubeIDE 1.19.0, and there have been some changes since the original post was written.

As @IRakh.1 has found, the .ioc settings for BLE no longer include CFG_BLE_LSE_SOURCE. However, this is no longer necessary because apparently now the auto-generation code looks at your clock configuration and autogenerates the change in app_conf.h to add SHCI_C2_BLE_INIT_CFG_BLE_LS_CLK_HSE_1024 in place of SHCI_C2_BLE_INIT_CFG_BLE_LS_CLK_LSE in the CFG_BLE_LS_SOURCE.

Once I did steps 1, 2, 3a, 3c, 4, and 5, I was up and running without the RTC crystal. Then I noticed I still needed to do steps 6 and 7. I did those steps also, and everything still worked. Bottom line, don't worry about step 3b and the change that @gongw mentioned in 2022 is now taken care of automatically.

0 Kudos

Feedback

Top

About STMicroelectronics

Who we are

Investor relations

Sustainability

Innovation & technology

Careers

Blog

General terms and conditions

Connect with us

Contact ST offices

Find sales offices & distributors

Community

Newsroom

Events & trainings

Browse

Shortcuts

Sitemap

Compliance, ethics & privacy

Ethics and compliance

ST ethics hotline

Privacy portal

Feedback

Follow us

All rights reserved © 2025 STMicroelectronics Terms of use Sales terms & conditions Trademarks Privacy portal Manage cookies

Feedback